Context-Conditioned Causal Alignment for Scientific Literature: Separating Cross-Domain Analogies, True Contradictions, and Contextual Heterogeneity

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***Abstract*—Scientific papers can describe the same mechanism using different disciplinary vocabularies, and they can appear to disagree while studying different populations, interventions, environments, or time horizons. Existing embedding retrieval emphasizes semantic proximity, while current graph-comparison and contradiction systems do not jointly align mechanism structure, relation polarity, and scope conditions. We propose Context-Conditioned Signed Partial Transport (CCST), a framework for aligning variable-size reported-mechanism graphs. Each graph preserves population or system, intervention, mediating steps, outcome, time, effect direction, epistemic status, uncertainty, and source provenance. CCST permits irrelevant mechanisms to remain unmatched, compares typed signed relations, and gates contradiction decisions on context compatibility. It assigns agreement, cross-domain analogy, contradiction candidate, contextual heterogeneity, or insufficient comparability and returns a minimal, evidence-linked disagreement certificate. Three sparse indexes reduce exact comparisons from all pairs to an empirical $\mathcal{O}(N \log N + Nk_{cccsr})$ pipeline. A 2,000-document public corpus and a four-stratum benchmark will test relation accuracy, false-contradiction reduction, cross-domain retrieval, certificate faithfulness, privacy-utility trade-offs, and scaling.**

***Index Terms*—scientific literature mining, causal-claim alignment, cross-domain analogy, contradiction detection, partial optimal transport, privacy-aware NLP**

I. PROBLEM DEFINITION

Let $\mathcal{D} = \{d_1, ..., d_N\}$ be a corpus of scientific documents. The goal is not to infer causal truth from prose, but to compare reported mechanisms: what each document claims happens, under which conditions, with what direction and uncertainty, and on what textual evidence. A document embedding is useful for topical search but can (i) miss structurally similar mechanisms written in disjoint vocabularies and (ii) make context-shifted outcomes look like either agreement or contradiction. A valid contradiction decision therefore requires three stages: align corresponding mechanisms, verify compatible material conditions, and only then locate an opposing signed relation or mutually exclusive assertion.

The research question is: Can reported mechanisms be aligned in a way that is tolerant to domain terminology but sensitive to direction, polarity, and scope, so that genuine contradiction candidates are separated from transferable analogies and context-dependent outcomes at corpus scale? The system produces five relations: agreement; cross-domain analogy; contradiction candidate; contextual heterogeneity; and insufficient comparability. "Candidate" is deliberate: structural alignment does not establish that either paper is causally correct.

The novelty is not the combination of embeddings, graphs, language models, clustering, and privacy controls. Those capabilities already exist. The proposed contributions are (1) a context-gated signed partial-transport objective, (2) a five-way relation definition that separates contradiction from heterogeneity, (3) a provenance-grounded minimal disagreement certificate, and (4) a controlled and naturally annotated benchmark for these distinctions.

II. DATA SETS

Source corpus. The course-scale study will freeze a 2,000-document public corpus rather than the earlier 10,000-document target. Title-abstract records will be sampled from PubMed and arXiv across biomedical, computer-security, control-systems, and related scientific categories. Metadata and source identifiers will be retained for reproducibility. Duplicates, retracted records when marked, records without abstracts, and non-English records will be excluded. The public corpus is used for extraction, retrieval, and empirical scaling; restricted organizational documents are outside the course-project scope.

Four-stratum benchmark. We target 240 evaluated pairs, balanced across: (1) expert-specified mechanism graphs rendered with controlled changes to vocabulary, sign, context, time, or irrelevant mechanisms; (2) natural cross-domain analogies; (3) scientific contradiction, agreement, and contextual-heterogeneity pairs; and (4) adversarial hard negatives such as quoted-but-not-endorsed claims, simulations versus observations, missing context, and lexically similar but structurally different mechanisms. Controlled transformations provide objective node, edge, and relation labels by construction. Natural pairs are independently labeled by at least three annotators using a common schema; agreement is reported with Krippendorff's α before adjudication.

Documents and transformation families are grouped before a 60/20/20 train-development-test split so that paraphrases, templates, or source documents cannot cross partitions. Development data set thresholds and calibration; the held-out test set is opened only after the pipeline is frozen. All evidence spans remain linked to their source. Privacy experiments use controlled masking of public text, not confidential records, and report both leakage and utility.

III. STATE-OF-THE-ART METHODS AND ALGORITHMS

Analogical-search systems already retrieve purpose- or mechanism-related papers across domains [1]. Document-level causal extraction already represents events in graphs and hypergraphs [2]. Fused Gromov-Wasserstein (FGW) compares attributed graphs [3]; Fused Network GW adds edge features [4]; and LinearFGW reduces comparison cost [5]. However, these methods do not by themselves establish context-compatible contradiction or produce a localized evidence certificate. Biomedical studies further show that species, dose, population, and other contextual variables explain many apparent conflicts [6]. Curvature-based clustering [7] and LLM supervised document-graph clustering [8] can help prioritize or refine graph structure, but neither curvature nor an LLM judgment is treated as ground truth.

Table I makes the required classical-versus-modern comparison explicit. Classical data-mining and ML methods remain evaluation baselines; modern models supply semantic representations and structured extraction; CCST is compared with both rather than credited for merely combining them.

IV. RESEARCH PLAN

A. Contextualized Reported-Mechanism Graphs

A document may contain several mechanisms and competing results; it will not be collapsed to one subject-verb-object triple. Claim-bearing spans are segmented, and each mechanism is represented as

$$q_{im} = (C, P, A, M, Y, \tau, \delta, \eta, u, p)_{im}, \tag{1}$$

where C is context; P population or system; A intervention or trigger; M one or more mediating steps; Y outcome; τ time; δ effect direction or magnitude; η epistemic status; u uncertainty and p source provenance. A document becomes a variable-size graph $G_i = (V_i, E_i, X_i, R_i, Z_i, p_i, \rho_i)$ with node roles, confidence-weighted mass, and edge type, direction, sign, time lag, and confidence. Up to M_{max} mechanisms are retained for coverage and evidential importance; overflow is processed in linked chunks. Missing context is marked unknown rather than assumed compatible.

B. Normalization, Sparse Retrieval, and Privacy Gate

Role normalization permits comparison across fields, while value compatibility keeps "adult patients," "mice," and "router nodes" distinct. For each material context dimension h, $c_h(i,j) \in [0,1]$ records compatible, partly compatible, incompatible, or unknown conditions. The extractor runs locally, retains raw passages only in the authorized environment, and can replace identifiers with pair-consistent typed placeholders. External review, if permitted, receives only the minimal aligned subgraph after leakage checks; otherwise the system uses a local model, authorized human review, or abstention.

All-pairs CCST would require $\mathcal{O}(N^2)$ alignments. We instead build semantic, unsigned-mechanism, and context sketches and retrieve k_s, k_m, and k_c approximate neighbors. Analogy search permits semantic distance but requires mechanism similarity; contradiction search requires both mechanism similarity and plausible context compatibility. A small diversity sample estimates missed-neighbor risk. With bounded graph size, the intended measured cost is $\mathcal{O}(N \log N + N k_{cccst})$, not a worst-case guarantee; candidate recall against exact sampled subsets will be reported.

C. Context-Conditioned Signed Partial Transport

For confidence masses p_i, p_j, a partial coupling $T \geq 0$ satisfies $T\mathbf{1} \leq p_i$ and $T^\top \mathbf{1} \leq p_j$. Let $U(T) = ||p_i||_1 + ||p_j||_1 - 2\mathbf{1}^\top T \mathbf{1}$ denote unmatched mass and let $L_{uv,u'v'}$ compare two typed edges. CCST minimizes

$$D_{CCST}(G_i, G_j) = \min_T \Bigg[\sum_{u,u'} M_{uu'} T_{uu'} + \alpha \sum_{u,v,u',v'} L_{uv,u'v'} T_{uu'} T_{vv'} + \lambda_u U(T) + \lambda_g \Phi_{gate} \Bigg] \tag{2}$$

M compares attributes, roles, and context; L compares edge type, direction, sign, temporal lag, and confidence; U avoids forced correspondences; and Φ_{gate} prevents a low structural cost from becoming contradiction evidence when scope conditions are incompatible or unknown. An unsigned view ignores relation sign to detect mechanism shape, while the signed view localizes polarity changes.

D. Relation, Certificate, and Discovery Graph

A calibrated classifier uses the alignment, context compatibility, sign-flip mass, domain separation, and extraction uncertainty to assign the five relations and abstains above a prespecified error-risk threshold. Every non-abstained decision returns a certificate containing aligned nodes and edges, matched/incompatible/unknown context fields, the smallest high-confidence disagreement set, unmatched mechanisms, and supporting passages. Certificate elements are added by contribution until they explain a preregistered fraction of the signed-versus-unsigned score gap.

The corpus graph stores a five-probability relation vector on each retained edge rather than one positive similarity. Agreement and analogy support mechanism families; contradiction becomes a signed negative constraint; heterogeneity remains an explanatory link; and insufficient cases are retained for review. Signed correlation clustering is compared with ordinary spectral clustering and Leiden-style baselines. Optional curvature and LLM review are budgeted acquisition features, not contradiction evidence.

TABLE I
CLASSICAL, MODERN, AND PROPOSED METHODS

Stage	Classical DM/ML	Modern AI/DL	This project
Retrieval	BM25, TF-IDF, LDA	SBERT, Contriever	Semantic, mechanism, and context ANN indexes
Grouping	K-Means, spectral clustering	LLM-supervised graph clustering	Signed multi-relation communities
Extraction	Rules and dependency SVO	Transformer or local-LLM extraction	Typed, multi-mechanism graph with provenance
Comparison	Graph edit, role matching, FGW/LinearFGW	NLI and direct LLM pair judgment	Context-gated, signed partial transport
Decision	Logistic regression and fixed thresholds	Calibrated neural classifier	Abstention plus disagreement certificate

V. EVALUATION PLAN

Baselines. Classical baselines are BM25 [11], TF-IDF/LDA [12] with K-Means, logistic regression on hand-crafted alignment features, graph edit or role-constrained matching, and ordinary FGW/LinearFGW. Modern baselines are SBERT [9], Contriever [10], transformer NLI, direct local-LLM pair judgment, and LLM-supervised graph clustering. Additional ablations remove partial matching, signed edges, each context field, provenance, each candidate index, calibration, and certificates.

Metrics. Extraction is evaluated with field and evidence-span F1. Alignment uses node/edge correspondence F1, sign-flip localization, and unmatched-mechanism accuracy. Relation evaluation reports five-class macro-F1, per-class precision/recall, calibration error, abstention coverage, and the critical false-contradiction rate on context-shift hard negatives. Retrieval uses MRR, Recall@k, and nDCG. Certificates use cited-relation precision/recall, evidence-span intersection-over-union, deletion faithfulness, minimality, and blinded expert usefulness. Efficiency reports candidate recall relative to exact sampled search, runtime, peak memory, exact-alignment count, review calls, and performance versus budget. Privacy reports identifier retention, linkage success, blocked-record rate, relation accuracy, and the privacy-utility frontier.

All final comparisons use the untouched test partition, five random seeds where stochastic training is involved, bootstrap 95% confidence intervals, and paired significance tests. Results are stratified by lexical overlap, domain distance, graph size, context completeness, and relation type. Success requires CCST to improve macro-F1 and cross-domain Recall@k over the strongest baseline while materially reducing false contradictions without doing so only through excessive abstention.

VI. FEASIBILITY, RISKS, AND EXPECTED OUTPUTS

Feasibility. The two-month implementation is deliberately smaller than the full research agenda. It uses 2,000 public documents, bounded mechanism graphs, exact alignment only on stratified reference subsets, and approximate retrieval elsewhere. The required deliverable is a reproducible offline prototype containing extraction, the three candidate indexes, CCST, calibrated five-way classification, certificates, and the stated baselines. Curvature-guided acquisition, an extensive expert-utility study, and a polished interactive interface are stretch goals and will not displace the core experiments.

Risks and controls. Extraction errors can propagate into a seemingly precise alignment, so extraction and downstream errors are reported jointly and every graph field retains evidence. No fixed context schema can capture every scientific condition; the schema therefore includes missingness and an ”other material condition” field, and unsupported cases must abstain. Approximate indexes can miss rare analogies, so candidate recall is measured against exact search on smaller subsets and a diversity sample is reserved. A sign difference alone may reflect measurement, design, uncertainty, or epistemic status rather than contradiction; these factors enter the benchmark and certificate. Language-model review is never treated as ground truth, and sensitive material cannot be sent outside its approved handling environment.

Expected outputs. The project will deliver versioned source records, a documented graph schema, controlled transformations and independently annotated natural pairs, baseline and CCST implementations, fixed evaluation scripts, and a final error analysis. The scientific output is correspondingly narrow: evidence that context-gated partial alignment can improve retrieval and reduce false contradiction claims. The system will not claim universal causal discovery, perfect privacy, or a worst-case subquadratic guarantee.

VII. PROJECT TIMELINE: TASKS, DESCRIPTIONS, DEADLINES

TABLE II
EIGHT-WEEK COURSE PROJECT SCHEDULE

Task	Deliverable	Deadline
Scope and schema	Freeze question, graph schema, privacy rules, and annotation guide	Sept. 30
Corpus and baselines	Freeze 2,000 documents; implement BM25/TF-IDF/LDA and embedding retrieval	Oct. 9
Extraction and benchmark	Provenance-linked extractor; controlled cases; first independent annotations	Oct. 16
CCST prototype	Partial solver, signed/unsigned views, synthetic property tests	Oct. 23
Sparse pipeline	Three indexes, calibrated relation model, abstention, certificates	Oct. 31
Main experiments	Baselines, held-out evaluation, candidate-recall and scaling results	Nov. 6
Ablations and audit	Context/sign/index ablations, error analysis, privacy-utility test	Nov. 13
Final integration	Reproducible code, figures, report, and presentation	Nov. 23

VIII. DIVISION OF WORK

TABLE III
PRIMARY OWNERSHIP; ALL MEMBERS REVIEW AND WRITE

Member	Primary responsibilities
Shravan Krishna Vijay	Corpus acquisition, preprocessing, BM25/TF-IDF/LDA baselines, reproducibility
Sri Harshith Goli	CCST formulation and solver, signed/unsigned tests, method integration
Subhash Shivaprasad	Multi-mechanism extraction, provenance, context schema, privacy gate
Tarun Kumar Bantupalli	Sparse indexes, scaling measurements, calibration and budgeted review
Vedang Vasant Avaghade	Benchmark coordination, statistical evaluation, certificates and result presentation
All members	Independent annotation, code review, error analysis, report editing, and final presentation

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AI USAGE DECLARATION

AI assistance disclosure. This report was drafted and formatted with AI assistance. Tools used: ChatGPT (GPT-5.6 Sol) for the initial draft, and Claude (Claude Sonnet 5.5, Anthropic) to revise the wording for a more natural tone, formatting, and converting the document to LaTeX. We verified all the references and corrected them where they differed. I reviewed and edited the final text, and I am responsible for its content.